On the Existence of a Solution to the Coupled HJB–KFP System

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1 Introduction

Context and Motivation Mean field games is a branch of game theory devoted to the analysis of a special subset of 'games' within the discipline. These systems or 'games' are distinguished by having a very large number of players, each of whom is considered almost insignificant with minimal influence on the system as a whole. In measure theory this corresponds to nullifying sets with countable measure and thinking of functions in the distributional sense. Furthemore, each player is a rational observer and the general trend around them will inform their decision on how to act, with the aim of acheiving the most optimal outcome. The famous example in economics of such a game, introduced by Aumann, is the behaviour of stock traders whom we treat as a continuum [1]. In this paper we consider the equilibrium problem characterised by a coupled pair of equations. These are a Hamilton–Jacobi–Bellman (HJB) equation with a Kolmogorov–Fokker–Planck (KFP) equation

$$\begin{aligned} -\Delta u + H(x, \nabla u) + \lambda u &= f(x, m) \quad \text{in } \Omega \text{ (HJB)}, \\ -\Delta m - \operatorname{div}\big(m \frac{\partial H}{\partial p}(x, \nabla u)\big) + \lambda m &= G(x) \qquad \text{in } \Omega \text{ (KFP)}, \end{aligned} \tag{1.1}$$

together with homogeneous Dirichlet boundary Assumptions $u = 0$ and $m = 0$ on $\partial\Omega$.

Interpretation of Equations $m(x)$ represents the player density and $u(x)$ the value function of a representative agent in the mean field game. In other words, $u(x)$ is a player's optimal solution achieved by minimising the expected cost at position x. Here $\frac{\partial H}{\partial p}$ denotes the partial derivative of H with respect to the momentum variable (i.e. the ∇u argument), and $\lambda > 0$.

(HJB) describes a representative player's optimal control behaviour for a given player density, m, in the game. On the other hand, (KFP) takes the macroscopic view and gives us the evolution of $m(\cdot)$ when every individual player satisfies the optimal solution u.

Each player tries to minimise their cost function $f(x, m)$ (in HJB) and the optimal control for an individual player is given by $\alpha^* = -\frac{\partial H}{\partial p}(x, \nabla u)$ which we can see makes up the (KFP). The Δ terms are called 'diffusion' terms which account for the randomness in the game (determined by Brownian motion) and prevent the perfect alignment of players even if they all adopt the same strategy. Lastly, $G()$ is an exogenous function, controlling the entry /exit of players to and from the system and as such will directly affect m which is why it appears in the (KFP). [4, 2]

Mathematical Contribution In this paper we analyse this coupled system of equations and in particular we investigate the existence of a weak solution pair (u, m) in $H_0^1(\Omega) \times H_0^1(\Omega)$. In addition, we consider a more general formulation of the system and prove an existence result in this extended setting, which constitutes an original contribution of this work.

Assumptions

1. Ω is bounded and $\partial\Omega$ is smooth.
2. $H(x, \nabla u) : \mathbb{R}^{n+1} \to \mathbb{R}$ is
   - Lipschitz continuous with respect to ∇u with Lipschitz constant $\tilde{C}$,
   - bounded by $C\big(|\nabla u| + 1\big)$ for a given constant C, for all $x \in \mathbb{R}$ and $\nabla u \in \mathbb{R}^n$.
3. The partial derivative H_p is continuous with respect to ∇u and essentially bounded by some constant L.
4. λ is sufficiently large (in particular $\lambda > \max(\frac{L^2}{4}, \frac{C^2}{2}, \frac{\tilde{C}^2}{4})$).
5. $f(x, m(x)) : \Omega \times L^2(\Omega) \to L^2(\Omega)$ and is continuous w.r.t. m.
6. $G(x) : \Omega \to L^2(\Omega)$.

Example 1. A Hamiltonian satisfying Assumption 2 is

$$H(x, p) = p + V(x) \text{ on } \Omega = (0, 1),$$

where $V \in L^\infty(\Omega)$. In this case, the associated drift, $\frac{\partial H}{\partial p}$, is constant.

Example 2. For a coupling term satisfying Assumption 3, one may consider

$$f(x, m) = a(x)\,\phi(m),$$

where $a(x) \in L^\infty(\Omega)$ and $\phi \in L^2(\Omega)$ is continuous.

Main Result

Theorem 1.1 (Existence of the MFG solution)**.** *Let the Mean Field Game system be characterised by 1.1 and suppose the assumptions above are true. Then a solution* $(u, m) \in H^1_0(\Omega) \times H^1_0(\Omega)$ *exists.*

We will prove this theorem throughout the rest of this paper.

2 Method

Our proof consists of two stages. First, we demonstrate solvability for each equation separately, while fixing the other variable (HJB with fixed m, and KFP with fixed u). Second, we show the existence of a solution *pair* solving both equations simultaneously. Before starting our proof it's important to introduce the following inequalities. These are essential and will lend themselves to almost every proof throughout this paper.

Definition 2.1. p, $q \in [1, \infty]$ are *conjugate parameters* if $\frac{1}{q} + \frac{1}{p} = 1$

Theorem 2.2 (Hölder's inequality). *Let $f \in L^p(\Omega)$ and $g \in L^q(\Omega)$ where p, q are conjugate parameters. Then*

$$\|fg\|_1 \leq \|f\|_p \, \|g\|_q$$

Remark 2.3. In particular, when p, $q = 2$ we get the Cauchy–Schwarz inequality

$$\|fg\|_1 \leq \|f\|_2 \, \|g\|_2$$

Theorem 2.4 (Young's inequality). *Let a, $b \in \mathbb{R}$ then*

$$2ab \leq \frac{a^2}{\varepsilon^2} + \varepsilon^2 b^2 \qquad \forall \varepsilon > 0$$

2.1 Independent unique solutions

2.1.1 Solving Kolmogorov–Fokker–Planck

To solve

$$-\Delta m - \operatorname{div}(m \frac{\partial H}{\partial p}(x, \nabla u)) + \lambda m = G(x)$$

with fixed u, we note that the equation is linear in $m(x)$, so we appeal to the Lax–Milgram theorem [5].

Theorem 2.5 (Lax–Milgram). *Let V be a Hilbert space and let $a : V \times V \to \mathbb{R}$ be a bilinear form such that for all $u, v \in V$,*

(i) (boundedness) $|a(u, v)| \leq c_1 \|u\| \, \|v\|$,

(ii) (coercivity) $a(v, v) \geq c_0 \|v\|^2$.

Let $l : V \to \mathbb{R}$ be a bounded linear functional. Then there exists a unique $u \in V$ such that

$$a(u, v) = l(v) \qquad \text{for all } v \in V.$$

Moreover, $\|u\|_V \leq \frac{1}{c_0} \|l\|_{V^*}$.

For our purposes, we take $a(m, v) = l(v)$ for all $v \in H_0^1(\Omega)$ to be the weak formulation of (KFP), i.e.

$$a(m, v) = \int_\Omega \nabla m \cdot \nabla v + m \, \nabla v \cdot \frac{\partial H}{\partial p}(x, \nabla u) + \lambda m v \, dx, \qquad l(v) = \int_\Omega G(x) \, v \, dx.$$

We now go about showing that (KFP) meets the conditions for 2.5.

Proposition 2.6 (Boundedness of (KFP)). *The form $a(m, v)$ is bounded on $H_0^1(\Omega) \times H_0^1(\Omega)$.*

Proof. For arbitrary $m, v \in H_0^1(\Omega)$,

$$\begin{aligned} |a(m, v)| &\leq \int_\Omega |\nabla m||\nabla v| + |\nabla v||m||\frac{\partial H}{\partial p}| + \lambda |m||v| \, dx \\ &\leq \|\nabla m\|_{L^2} \|\nabla v\|_{L^2} + L \|m\|_{L^2} \|\nabla v\|_{L^2} + \lambda \|m\|_{L^2} \|v\|_{L^2} \end{aligned}$$

By the Cauchy–Schwarz inequality and Assumption 3. Therefore,

$$|a(m, v)| \leq C \|m\|_{H^1} \|v\|_{H^1}.$$

Thus, $a(m, v)$ is bounded. $\square$

Proposition 2.7 (Coercivity of (KFP))**.** *If $\lambda > 0$ is sufficiently large, then $a(m, v)$ is coercive.*

Proof. Consider

$$\begin{aligned} a(v, v) &= \int_\Omega \nabla v \cdot \nabla v + v \nabla v \cdot \frac{\partial H}{\partial p}(x, \nabla u) + \lambda v^2 \, dx \\ &\geq \|\nabla v\|_{L^2}^2 - L \|v\|_{L^2} \|\nabla v\|_{L^2} + \lambda \|v\|_{L^2}^2 \end{aligned}$$

By Assumption 3. Now applying Young's inequality we get

$$a(v, v) \geq \Big(1 - \frac{\varepsilon^2}{2}\Big) \|\nabla v\|_{L^2}^2 + \Big(\lambda - \frac{L^2}{2\varepsilon^2}\Big) \|v\|_{L^2}^2, \qquad \forall \varepsilon \in \mathbb{R},$$

To ensure $1 - \varepsilon^2/2 > 0$, impose $\varepsilon^2 < 2$. Choosing $\varepsilon^2 = 2$ (as a limiting choice) gives the weakest requirement

$$\lambda > \frac{L^2}{4},$$

which ensures coercivity. $\square$

Thus $a(\cdot, \cdot)$ is bounded and coercive.

Theorem 2.8 (Existence of unique (KFP) solution)**.** *Let the Kolmogorov-Fokker-Planck equation be as in 1.1 and suppose that $\lambda > 0$ is sufficiently large. Then for any fixed $u \in H_0^1(\Omega)$, there exists a unique $m \in H_0^1(\Omega)$ solving the equation.*

Proof. Let $a(m, v) = l(v)$ be the weak formulation of the (KFP) equation as defined above 2.1.1. By 2.6 and 2.7 we know that $a(\cdot, \cdot)$ is bounded and coercive. Left to show is that $l(v)$ is bounded. Consider

$$\|l\| = \sup_{\|v\|_{H^1} \leq 1} \left| \int_\Omega G v \, dx \right| \leq \sup_{\|v\|_{L^2} \leq 1} \|v\|_{L^2} \|G\|_{L^2}.$$

By Cauchy-Schwarz. Therefore,

$$\|l\| \leq \|G\|_{L^2(\Omega)}.$$

Thus, l is bounded. By the Lax-Milgram Theorem 2.5, there exists a unique solution $m \in H_0^1(\Omega)$ solving the equation. $\square$

Remark 2.9. Lax-Milgram conveniently applies here only because (KFP) is linear in m once $u \in H_0^1(\Omega)$ is fixed.

2.1.2 Solving Hamilton–Jacobi–Bellman

Lax–Milgram does not apply directly to

$$-\Delta u + H(x, \nabla u) + \lambda u = f(x, m)$$

with fixed m, because the equation is non-linear so we resort to a fixed point method instead. The idea, is to create some kind of map from (HJB) with the right properties such that it has a fixed point and for that very fixed point to be our solution. Remarkably, this fixed point method is general enough that we can use it to solve for existence in the system as whole. Indeed, this will be the concern of the next subsection. Throughout this section, X denotes a real Banach space. We now introduce the fixed point theorems that rely on suitable properties for the (HJB) context.

Theorem 2.10 (Schauder's fixed point theorem)**.** *If $K \subset X$ is compact and convex, and $A : K \to K$ is continuous, then A has a fixed point in K.*

Definition 2.11. A mapping $A : X \to X$ is called *compact* if for every bounded sequence $(u_k)_{k=1}^{\infty}$, the sequence $(A[u_k])_{k=1}^{\infty}$ is precompact in X.

Theorem 2.12 (Schaefer's fixed point theorem)**.** *Suppose $A : X \to X$ is continuous and compact, and assume the set*

$$\{u \in X : u = \mu A[u] \text{ for some } 0 \leq \mu \leq 1\}$$

is bounded. Then A has a fixed point. [1]

Let us now design the map whose fixed point is a solution to (HJB). Consider

$$A : H_0^1(\Omega) \to H_0^1(\Omega), \qquad A[u] = w \quad \text{where} \quad -\Delta w + \lambda w = f(x, m) - H(x, \nabla u).$$

We know w exists by Lax–Milgram 2.5, so A is well defined. Furthermore by regularity theory [2] we have

$$\begin{aligned}
\|w\|_{H^2(\Omega)} &\leq C \|f(x, m) - H(x, \nabla u)\|_{L^2(\Omega)} \\
&\leq C \|f(x, m)\|_{L^2(\Omega)} + C \|H(x, \nabla u)\|_{L^2(\Omega)} \\
&\leq C \|f(x, m)\|_{L^2(\Omega)} + \hat{C} \||\nabla u| + 1\|_{L^2(\Omega)} \qquad \text{(Condition 2)}
\end{aligned}$$

[1] See [3, Section 9.2] for proofs.
[2] See [3, Section 6.3]

Where C, $\hat{C}$ are constants - note that unless explicitly mentioned, these constants are not the same as the ones appearing in our assumptions above. We now prove the existence of an (HJB) solution by way of our fixed point map.

Theorem 2.13 (Existence of (HJB) Solution)**.** *If $\lambda > 0$ is sufficiently large, then for any fixed $m \in H_0^1(\Omega)$ there exists $u \in H^2(\Omega) \cap H_0^1(\Omega)$ solving (HJB).*

Proof. [3] In other words, for sufficiently large $\lambda > 0$, A has a fixed point. We show first that A meets the right criteria for Schaefer's Theorem. We assert that A is continuous and compact.
If $u_k \to u$ in $H_0^1(\Omega)$ and we denote $w_k = A[u_k]$ then higher regularity tells us

$$\sup_k \|w_k\|_{H^2(\Omega)} < \infty$$

(i.e (w_k) is a bounded sequence in $H^2(\Omega) \cap H_0^1(\Omega)$) and there exists a convergent subsequence $(w_{k_j})_{j=1}^{\infty} \to w$ in $H_0^1(\Omega)$.
Now the weak form of (HJB) with w_{k_j} is

$$\int_\Omega \nabla w_{k_j} \cdot v + \lambda w_{k_j} \cdot v \, dx = \int_\Omega (f(x,m) - H(x, \nabla u_{k_j})) \cdot v \, dx, \text{ for any } v \in H_0^1(\Omega)$$

and consequently

$$\int_\Omega \nabla w \cdot v + \lambda w \cdot v \, dx = \int_\Omega (f(x,m) - H(x, \nabla u)) \cdot v \, dx, \text{ for any } v \in H_0^1(\Omega)$$

by Lipschitz continuity of H (Condition 2). Thus, $w = A[u]$. Hence, $A[u_k] \to A[u]$ in $H_0^1(\Omega)$, so A is continuous. A is compact by similar reasoning since a bounded sequence $(u_k)_{k=1}^{\infty}$ yields a bounded sequence $(A[u_k])_{k=1}^{\infty}$ in $H^2(\Omega)$ which has a strongly convergent subsequence in $H_0^1(\Omega)$.

It remains to show

$$\{u \in H_0^1(\Omega) : u = \mu A[u] \text{ for some } 0 \leq \mu \leq 1\}$$

is bounded.

Substituting $w = u/\mu$ and setting $v = u$, we obtain

$$\begin{aligned}
\int_\Omega |\nabla u|^2 + \lambda |u|^2 \, dx &= \mu \int_\Omega fu - H(x, \nabla u) u \, dx \\
&\leq \int_\Omega |fu| + |H(x, \nabla u) u| \, dx \\
&\leq \int_\Omega |fu| + C(|\nabla u| + 1)|u| \, dx \qquad \text{(Assumption 2)}
\end{aligned}$$

[3] Proof is adapted from [3, Section 9.2]

To isolate $|\nabla u|^2$ and u^2 terms, we apply Young's inequality 2.4 twice over to give

$$\begin{aligned}\int_\Omega |\nabla u|^2 + \lambda |u|^2\,dx &\le \frac{1}{2}\int_\Omega \frac{f^2}{\varepsilon^2} + \varepsilon^2 u^2 + \frac{1}{\delta^2}|Cu|^2 + \delta^2(|\nabla u|+1)^2\,dx, \qquad \varepsilon\in(0,\infty),\ \delta\in(0,1)\\ &\le \frac{1}{2}\int_\Omega \frac{f^2}{\varepsilon^2} + \Big(\varepsilon^2+\frac{C^2}{\delta^2}\Big)u^2 + \delta^2(1+\gamma^2)|\nabla u|^2 + \delta^2(1+\frac{1}{\gamma^2})\,dx, \qquad \gamma\in(0,\infty).\end{aligned}$$

We require $\delta^2(1+\gamma^2)<1$ (in particular, $\delta^2<1$ is necessary). $\varepsilon,\gamma>0$ can be arbitrarily small, so provided $\lambda>\frac{C^2}{2}$ there exist sufficiently small $\varepsilon,\gamma>0$ and sufficiently large $\delta^2<1$ such that

$$\lambda > \frac{1}{2}(\varepsilon^2+\frac{C^2}{\delta^2}),\ \text{and}\ 1>\delta^2(1+\gamma^2)$$

hold simultaneously. Thus we obtain

$$\int_\Omega a|\nabla u|^2 + b|u|^2\,dx \le \int_\Omega \frac{f^2}{\varepsilon^2} + \delta^2(1+\frac{1}{\gamma^2})\,dx \le \text{constant}$$

with $a,b>0$ constants. Thus if $\lambda>0$ is sufficiently large, we have $\|u\|_{H_0^1(\Omega)}\le C$ for some constant C that does not depend on $0\le\mu\le 1$. □

Remark 2.14. We can always tune ε,δ and γ for given $\lambda>\frac{C^2}{2}$ to satisfy boundedness. This utilises the strictness of the inequality in the condition on λ.

Proposition 2.15. *If $\lambda>0$ is sufficiently large, then the fixed point u is unique.*

Proof. Suppose u_1,u_2 both solve (HJB) for a given $m\in H_0^1(\Omega)$. Taking the weak difference gives

$$\begin{aligned}\|\nabla(u_1-u_2)\|_{L^2}^2 + \lambda\|u_1-u_2\|_{L^2}^2 &\le \int_\Omega |u_1-u_2|\,|H_1-H_2|\,dx\\ &\le \tilde{C}\,\|u_1-u_2\|_{L^2}\,\|\nabla(u_1-u_2)\|_{L^2}\end{aligned}$$

using Assumption 2 and Cauchy–Schwarz. Hence, by Young's inequality,

$$\|\nabla(u_1-u_2)\|_{L^2}^2 + \lambda\|u_1-u_2\|_{L^2}^2 \le \frac{1}{2}(\frac{\tilde{C}^2}{\varepsilon^2}\,\|u_1-u_2\|_{L^2}^2 + \varepsilon^2\,\|\nabla(u_1-u_2)\|_{L^2}^2) \qquad \varepsilon\in(0,\sqrt{2}).$$

Thus by the same reasoning as when we proved existence,

$$\lambda > \frac{\tilde{C}^2}{4} \implies u_1=u_2 \text{ in } L^2(\Omega) \iff u_1=u_2 \text{ in } H_0^1(\Omega).$$

□

The following theorem summarises our results.

Theorem 2.16 (Existence of unique (HJB) solution)**.** *Let the Hamilton-Jacobi equation be as in 1.1 and suppose that $\lambda > 0$ is sufficiently large. Then for any fixed $m \in H_0^1(\Omega)$, there exists a unique $u \in H_0^1(\Omega)$ solving the equation.*

2.2 Solving the Coupled System of Equations

Having solved the two equations in 1.1 in isolation, we now look to solve them as a system. As stated above, the fixed point theorems will allow us to prove existence of a solution pair for the system as a whole. The challenge, is to define and prove the existence of a fixed point for a suitable map. Our work in the previous subsection allows us to define the following maps.

Definition 2.17. Define $S : H_0^1(\Omega) \to H_0^1(\Omega)$ by $S(u) = m$, where m solves (KFP) for given u.

Definition 2.18. Define $T : H_0^1(\Omega) \to H_0^1(\Omega)$ by $T(m) = u$, where u solves (HJB) for given m.

Definition 2.19. Let T and S be as in definitions 2.17 and 2.18. Define

$$A : H_0^1(\Omega) \to H_0^1(\Omega), \qquad A := S \circ T.$$

That is, A takes $m \in H_0^1(\Omega)$, converts it to $u = T(m)$ solving (HJB), and then produces $\widetilde{m} = S(u)$ solving (KFP). If A has a fixed point, then $m = \widetilde{m}$, and the coupled system has a solution pair.

We verify the hypotheses of Schaefer's theorem.

Lemma 2.20. *Let A be defined as in 2.19 and suppose that $\lambda > 0$ as it appears in 1.1 is sufficiently large. Then, A is continuous.*

Since A is simply a composition of S and T, it suffices to show they both continuous.

Proposition 2.21. *If $\lambda > 0$ is sufficiently large, T is continuous.*

Proof. Let $(m_k)_{k=1}^{\infty} \to m$ in $H_0^1(\Omega)$. Denote $u = T[m]$, $u_k = T[m_k]$ and $f_k = f(x, m_k)$. Taking the weak formulation of the difference in (HJB) gives

$$\begin{aligned}
\|\nabla(u-u_k)\|_{L^2}^2 + \lambda \|u-u_k\|_{L^2}^2 &\leq \int_\Omega |u-u_k|\,|H-H_k| + |u-u_k|\,|f-f_k|\,dx \\
&\leq \tilde{C}\,\|u-u_k\|_{L^2}\,\|\nabla(u-u_k)\|_{L^2} + \|u-u_k\|_{L^2}\,\|f-f_k\|_{L^2}\,.
\end{aligned}$$

By Cauchy-Schwarz and Assumption 2. Since f is $L^2(\Omega)$-continuous (Assumption 5), $\|f-f_k\|_{L^2} \to 0$. Also, from the energy estimate

$$\|\nabla u_k\|_{L^2}^2 + \lambda \|u_k\|_{L^2}^2 \leq \int_\Omega |u_k|\,|f_k| + |u_k|\,|H_k|\,dx.$$

Reapplying the boundedness argument from 2.13 and noting that by it's convegence to f, f_k is bounded, we get that (u_k) is bounded in $H_0^1(\Omega)$. Thus, $\|u - u_k\|_{L^2} \|f - f_k\|_{L^2} \to 0$. Therefore, in the limit of k we end up considering

$$\|\nabla(u - u_k)\|_{L^2}^2 + \lambda \|u - u_k\|_{L^2}^2 \leq \tilde{C} \|u - u_k\|_{L^2} \|\nabla(u - u_k)\|_{L^2}$$

for which $\lambda > 0$ being sufficiently large implies $u_k \to u$ i.e T is continuous. [4] □

Proposition 2.22. *If $\lambda > 0$ is sufficiently large, S is continuous.*

Proof. Let $(u_k)_{k=1}^{\infty} \to u$ in $H_0^1(\Omega)$. Denote $m = S[u]$, $m_k = S[u_k]$ and $\frac{\partial H}{\partial p}_k = \frac{\partial H}{\partial p}(x, \nabla u_k)$. Taking the weak formulation of the difference in (KFP) yields

$$\|\nabla(m - m_k)\|_{L^2}^2 + \lambda \|m - m_k\|_{L^2}^2 \leq \int_\Omega |\nabla(m - m_k)| \, |m \frac{\partial H}{\partial p} - m_k \frac{\partial H}{\partial p}_k| \, dx.$$

Define the correction factor $\varepsilon_k := \frac{\partial H}{\partial p} - \frac{\partial H}{\partial p}_k$. Note that since $\frac{\partial H}{\partial p}_k$ is continuous w.r.t. u_k we have $\lim_{k\to\infty} \varepsilon_k = 0$. Then,

$$\begin{aligned}
\|\nabla(m - m_k)\|_{L^2}^2 + \lambda \|m - m_k\|_{L^2}^2 &\leq \int_\Omega |\nabla(m - m_k)| \, |m \frac{\partial H}{\partial p} - m_k \frac{\partial H}{\partial p}| + |\nabla(m - m_k)| \, |m_k||\varepsilon_k| \, dx \\
&\leq L \int_\Omega |\nabla(m - m_k)| \, |m - m_k| \, dx + \int_\Omega |\nabla(m - m_k)| \, |m_k| \, |\varepsilon_k| \, dx.
\end{aligned}$$

m and m_k are bounded in $H_0^1(\Omega)$ so as k gets large $|\nabla(m - m_k)| \, |m_k||\varepsilon_k| \to 0$. Therefore, by the largeness of λ (see 2.7), we get $m_k \to m$ i.e S is continuous. □

Remark 2.23. The image of S is always bounded as a direct consequence of Lax-Milgram (2.5).

2.2.1 Compactness of A

Lemma 2.24. *Let A be defined as in 2.19 and suppose that $\lambda > 0$ as it appears in 1.1 is sufficiently large. Then, A is compact.*

Since S is continuous, it suffices to show that T is compact in $H_0^1(\Omega)$:

$$T \text{ compact} \implies A = S \circ T \text{ compact.}$$

We introduce the following theroems. These will allow us to conclude convergence of a subsequence in a bigger space if we are given a bounded sequence in a smaller space. The particular relationship that concerns us is the one between $L^2(\Omega)$ and $H_0^1(\Omega)$.

Definition 2.25. Let X and Y be Banach spaces with $X \subset Y$. We say that X is compactly embedded in Y, written $X \subset\subset Y$, provided:

[4] See the uniqueness argument above (2.15).

(i) $\|x\|_Y \leq C\|x\|_X$ for some constant C,

(ii) each bounded sequence in X is precompact in Y.

Theorem 2.26 (Rellich–Kondrachov compactness theorem)**.** *Assume U is a bounded open subset of $\mathbb{R}^n$ and ∂U is C^1. Suppose $1 < p < n$. Then*

$$W^{1,p}(U) \subset\subset L^q(U) \quad \textit{for each } 1 \leq q < p^*, \qquad p^* = \frac{np}{n-p}.$$

Remark 2.27. Observe $p^* > p$. In particular,

$$W^{1,p}(U) \subset\subset L^p(U) \quad \text{for all } 1 \leq p < n.$$

Example 3. The space we're interested in, is $H_0^1(\Omega)$. We know that $H_0^1(\Omega) \subset H^1(\Omega)$. Thus, by Rellich Kondrachov and Condition 1

$$H_0^1(\Omega) \subset\subset L^2(\Omega)$$

.

Proposition 2.28. *If $\lambda > 0$ is sufficiently large, T is compact.*

Proof. Let $(m_k)_{k=1}^\infty$ be a bounded sequence in $H_0^1(\Omega)$. Since Ω satisfies Assumption 1, there exists a subsequence $(m_{k_j})_{j=1}^\infty$ converging in $L^2(\Omega)$ to some $m \in L^2(\Omega)$. We want to show $T(m_{k_j}) \to T(m)$. Remember the weak form of $T(m) - T(m_{k_j})$

$$\left\|\nabla(u - u_{k_j})\right\|_{L^2}^2 + \lambda\left\|u - u_{k_j}\right\|_{L^2}^2 \leq \int_\Omega |u - u_{k_j}|\,|H - H_{k_j}| + |u - u_{k_j}|\,|f - f_{k_j}|\,dx$$

Because f is $L^2(\Omega)$-continuous (Assumption 5), $L^2(\Omega)$ convergence is sufficient, and the continuity argument for T applies with m_{k_j}. Thus T is compact, and hence A is compact. □

We have shown that A is continuous and compact. The last step before we can apply Schauder's theorem is to prove that if there were to exist a fixed point, it would be bounded.

2.2.2 Boundedness of the Fixed Point Set

Consider

$$\{m \in H_0^1(\Omega) : m = \mu A[m] \text{ for some } \mu \in [0,1]\}.$$

The corresponding system, with $A[m]$ substituted for m/μ, is

$$-\Delta u + H(x, \nabla u) + \lambda u = f(x, m), \qquad \frac{1}{\mu}\Big(-\Delta m - \operatorname{div}(m\,\frac{\partial H}{\partial p}(x, \nabla u)) + \lambda m\Big) = G(x).$$

Rearranging the latter and using the Lax–Milgram bound implies

$$\|m\|_{L^2} \leq \frac{1}{c_0}\mu\,\|G\|_{L^2} \leq \frac{1}{c_0}\,\|G\|_{L^2},$$

which is bounded in $L^2(\Omega)$ (Assumption 6). By Schaefer's theorem, A has a fixed point. We may conlclude our finding with the next theorem.

Proof of the Main Result

Proof of Theorem 1.1. Let A be as in 2.19. A is compact and continuous and has bounded fixed points (if they were to exist) by the above, 2.20 and 2.24. By Schaefer's theorem (2.12), A has a fixed point. Thus, a solution pair $(u, m) \in H_0^1(\Omega) \times H_0^1(\Omega)$ to the MFG sytem exists. □

2.3 Swapping Composition Order of T and S

In hindsight, we can see that the composition of T followed by S was convenient because it provided a natural bound for the fixed point. We relied on the the fact that the image of S is always bounded (2.5) independent of m. An equaly valid construction for A however, is the reverse composition

$$A := T \circ S.$$

Here too the existence of a fixed point proves 1.1 yet crucially, recall that T has no obvious bound. We note that bounded sets under S remain bounded so A retains its continuity and compactness. Therefore, we need only demonstrate boundedness of the fixed point set for us to use Schaefer's Theorem. Under this new construction consider

$$\{u \in H_0^1(\Omega) : u = \mu A[u] \text{ for some } \mu \in [0,1]\}.$$

The corresponding system, with $A[u]$ substituted for u/μ, is

$$-\frac{\Delta u}{\mu} + H(x, \frac{\nabla u}{\mu}) + \frac{\lambda u}{\mu} = f(x, m), \qquad -\Delta m - \operatorname{div}\big(m \frac{\partial H}{\partial p}(x, \nabla u)\big) + \lambda m = G(x).$$

We obtain the weak form of (HJB) with $v = u$.

$$\begin{aligned}
\int_\Omega |\nabla u|^2 + \lambda |u|^2 \, dx &= \mu \int_\Omega fu - H(x, \frac{\nabla u}{\mu}) u \, dx \\
&\leq \int_\Omega |fu| + \mu \, |H(x, \frac{\nabla u}{\mu}) u| \, dx \\
&\leq \int_\Omega |fu| + \mu \, C(|\frac{\nabla u}{\mu}| + 1)|u| \, dx \qquad \text{(Assumption 2)} \\
&= \int_\Omega |fu| + C(|\nabla u| + 1)|u| \, dx
\end{aligned}$$

This is the same form as the inequality in the proof of 2.13. There we arrived at the following bound

$$\int_\Omega |\nabla u|^2 + \lambda |u|^2 \, dx \leq \frac{1}{2} \int_\Omega \frac{f^2}{\varepsilon^2} + \Big(\varepsilon^2 + \frac{C^2}{\delta^2}\Big) u^2 + \delta^2 (1 + \gamma^2) |\nabla u|^2 + \delta^2 (1 + \frac{1}{\gamma^2}) \, dx,$$

where ε, γ and δ are fixed and λ is sufficiently large. Taking the difference of both sides and denoting the constant factor which is independent of m or u as K yields

$$\epsilon \int_\Omega |\nabla u|^2 + |u|^2 \, dx \leq \frac{1}{2\varepsilon^2} \int_\Omega f(x,m)^2 \, dx + K,$$

where ϵ is the minimum of both factors of u^2 and ∇u^2. Equivalently,

$$\epsilon \left\| u \right\|_{H_0^1(\Omega)} \leq \frac{1}{2\varepsilon^2} \left\| f(x,m) \right\|_{L^2(\Omega)} + K.$$

The bound of u must be independent of m so we introduce a growth condition on f.

Proposition 2.29. *Let $|f(x,m)| \leq (|m|+1)$ then $\{u \in H_0^1(\Omega) : u = \mu A[u]$ for some $\mu \in [0,1]\}$ is bounded.*

Proof. We have

$$\begin{aligned}
\epsilon \left\| u \right\|_{H_0^1(\Omega)} &\leq \frac{1}{2\varepsilon^2} \left\| f(x,m) \right\|_{L^2(\Omega)} + K \\
&\leq \frac{1}{2\varepsilon^2} (\left\| m \right\|_{L^2(\Omega)} + |\Omega|) + K \\
&\leq \frac{1}{2\varepsilon^2} (\left\| m \right\|_{H_0^1(\Omega)} + |\Omega|) + K \\
&\leq \frac{1}{2\varepsilon^2} (\frac{1}{c_0} \left\| G \right\|_{L^2(\Omega)} + |\Omega|) + K
\end{aligned}$$

by the Lax-Milgram bound (2.5). Thus the set is bounded. □

We summarise the analysis of this subsection with a lemma.

Lemma 2.30. *Let $A = T \circ S$ and suppose that $\lambda > 0$ as it appears in 1.1 is sufficiently large. Then,*

$$|f(x,m)| \leq (|m|+1) \implies A \text{ has a fixed point.}$$

Proof. A is compact and continuous since T is compact and continuous (2.28, 2.2) and S is bounded continuous (2.22). The set of fixed points is bounded by 2.29. Thus, there exists a fixed point by Schaefer's Theorem. □

3 Extension to Non-linear (KFP)

The following is an analysis on a variation of 1.1 with the $G(\cdot)$ in question being dependent on m. Our new system of equations reads

$$\begin{aligned}
-\Delta u + H(x, \nabla u) + \lambda u &= f(x,m) \quad \text{in } \Omega, \\
-\Delta m - \operatorname{div}(m \frac{\partial H}{\partial p}(x, \nabla u)) + \lambda m &= G(x,m) \quad \text{in } \Omega.
\end{aligned} \tag{3.1}$$

In this scenario we cannot apply the Lax-Milgram Theorem to (KFP). We will therefore adopt fixed point theorems to find a solution but as with $f(x, m(x))$ we must impose Assumptions on $G(x, m(x))$.

7. $G(x, m) : \Omega \times L^2(\Omega) \to L^2(\Omega)$ is
   - Lipschitz continuous with respect to m with Lipschitz constant $\tilde{B}$,
   - bounded by $B(|x| + |m| + 1)$ for a given constant B, for all $x \in \Omega$ and $m \in L^2(\Omega)$.
8. In addition to our previous constraint on λ, we have $\lambda > \max(B + \frac{L^2}{4}, \tilde{B} + \frac{L^2}{4})$

Theorem 3.1 (Existence of Non-linear KFP Solution)**.** *If $\lambda > 0$ is sufficiently large, there exists $m \in H^2(\Omega) \cap H_0^1(\Omega)$ solving (KFP) with fixed u.*

Proof. See 2.13 and [3, Chapter 9]. The only difference is in the thershold of λ which we'll now derive. As in 2.13 we want to show

$$\{m \in H_0^1(\Omega) : m = \mu A[m] \text{ for some } 0 \le \mu \le 1\}$$

is bounded. [5] Substituting $w = m/\mu$, we obtain

$$\begin{aligned}
\int_\Omega |\nabla m|^2 + \lambda |m|^2 \, dx &= \mu \int_\Omega G(x, m)m - m\nabla m \frac{\partial H}{\partial p} \, dx \\
&\le \int_\Omega |G(x, m)||m| + L|m||\nabla m| \, dx \\
&\le \int_\Omega B(|x| + |m| + 1)|m| + L|m||\nabla m| \, dx \qquad \text{(Assumption 7)} \\
&\le \int_\Omega B|m|^2 + \frac{1}{2}(\frac{L^2}{\varepsilon^2}|m|^2 + \varepsilon^2|\nabla m|^2) + K \, dx \qquad \varepsilon \in (0, \sqrt{2}) \\
&\le \int_\Omega (B + \frac{L^2}{2\varepsilon^2})|m|^2 + \frac{1}{2}(\varepsilon^2|\nabla m|^2) + K \, dx.
\end{aligned}$$

Once more using Young's inequality. This yields a bound on $\|m\|_{H_0^1(\Omega)}$ provided $\lambda > B + \frac{L^2}{4}$. □

Proposition 3.2. *If $\lambda > 0$ is sufficiently large, then the fixed point m is unique.*

Proof. Suppose m_1, m_2 both solve (HJB). Taking the difference gives

$$\begin{aligned}
\|\nabla(m_1 - m_2)\|_{L^2}^2 + \lambda \|m_1 - m_2\|_{L^2}^2 &\le \int_\Omega |\frac{\partial H}{\partial p}| \, |m_1 - m_2| \, |\nabla(m_1 - m_2)| + |G_1 - G_2| \, |m_1 - m_2| \, dx \\
&\le L \|m_1 - m_2\|_{L^2} \|\nabla(m_1 - m_2)\|_{L^2} + \tilde{B} \, \|m_1 - m_2\|_{L^2}^2
\end{aligned}$$

[5] $A(\cdot)$ here is the map defined for fixed u not to be confused with the composition of S and T.

using Assumptions 2, 7 and Cauchy–Schwarz. Hence, by our previous use of Young's inequality in 2.7,

$$\lambda - \tilde{B} > \frac{L^2}{4} \implies m_1 = m_2 \text{ in } H_0^1(\Omega).$$

□

Thus our definition of $S(\,\cdot\,)$ in 2.17 remains well defined for (KFP) in its new form. We have left to prove that S is still continuous after which our previous results will hold for compactness of A.

Continuity of S #2 Oncemore, let $(u_k)_{k=1}^{\infty} \to u$ in $H_0^1(\Omega)$. Denote $m = S[u]$, $m_k = S[u_k]$, $\frac{\partial H}{\partial p}_k = \frac{\partial H}{\partial p}(x, \nabla u_k)$ and $G_k = G(x, m_k)$. Taking the weak formulation of the difference in (KFP) yields

$$\begin{aligned}\|\nabla(m - m_k)\|_{L^2}^2 + \lambda \|m - m_k\|_{L^2}^2 &\leq \int_\Omega |\nabla(m - m_k)|\, |m \frac{\partial H}{\partial p} - m_k \frac{\partial H}{\partial p}_k| + |G - G_k|\, |m - m_k|\, dx \\ &\leq \int_\Omega |\nabla(m - m_k)|\, |m \frac{\partial H}{\partial p} - m_k \frac{\partial H}{\partial p}_k| + \tilde{B}|m - m_k|^2\, dx.\end{aligned}$$

By λ being sufficiently large to account for the $\tilde{B}$ term and following the same line of reasoning in **??** we conclude that S is continuous.

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